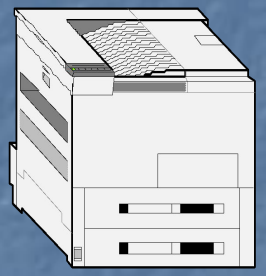
Overview of Data Communication and Computer Networks

Outline

- Data Communication Overview
  - Definition
  - Components
  - Protocols and standards
- Network Overview
  - Connectivity
  - Categories
  - Internetworks

Motivations

- Efficient way to share resources
  - Cost – less expensive
  - Accessibility – easier



- Efficient way to exchange information
  - Time – faster
  - Size – bigger
  - Correctness – more accurate

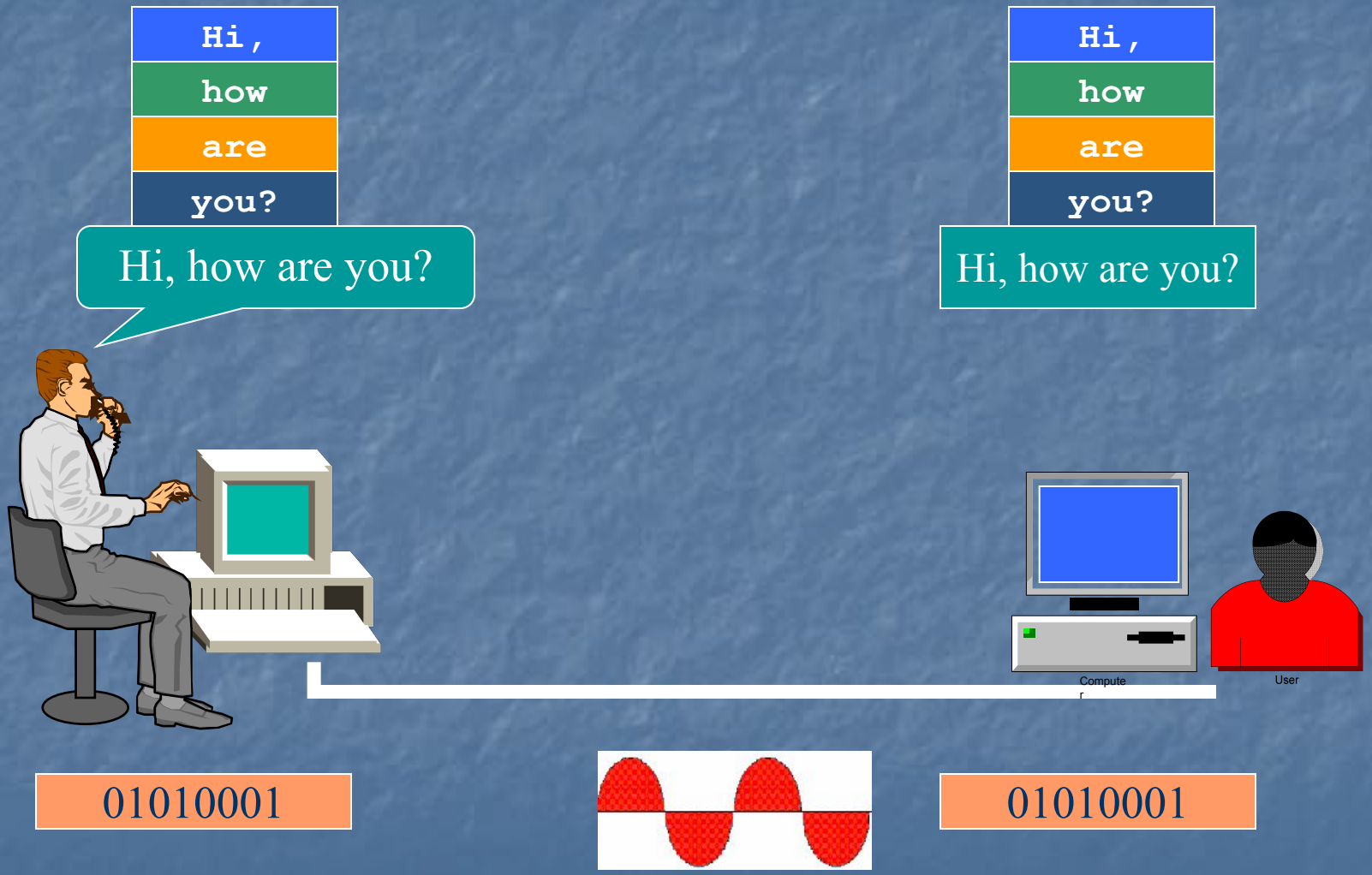


Data Communication: *Definition*

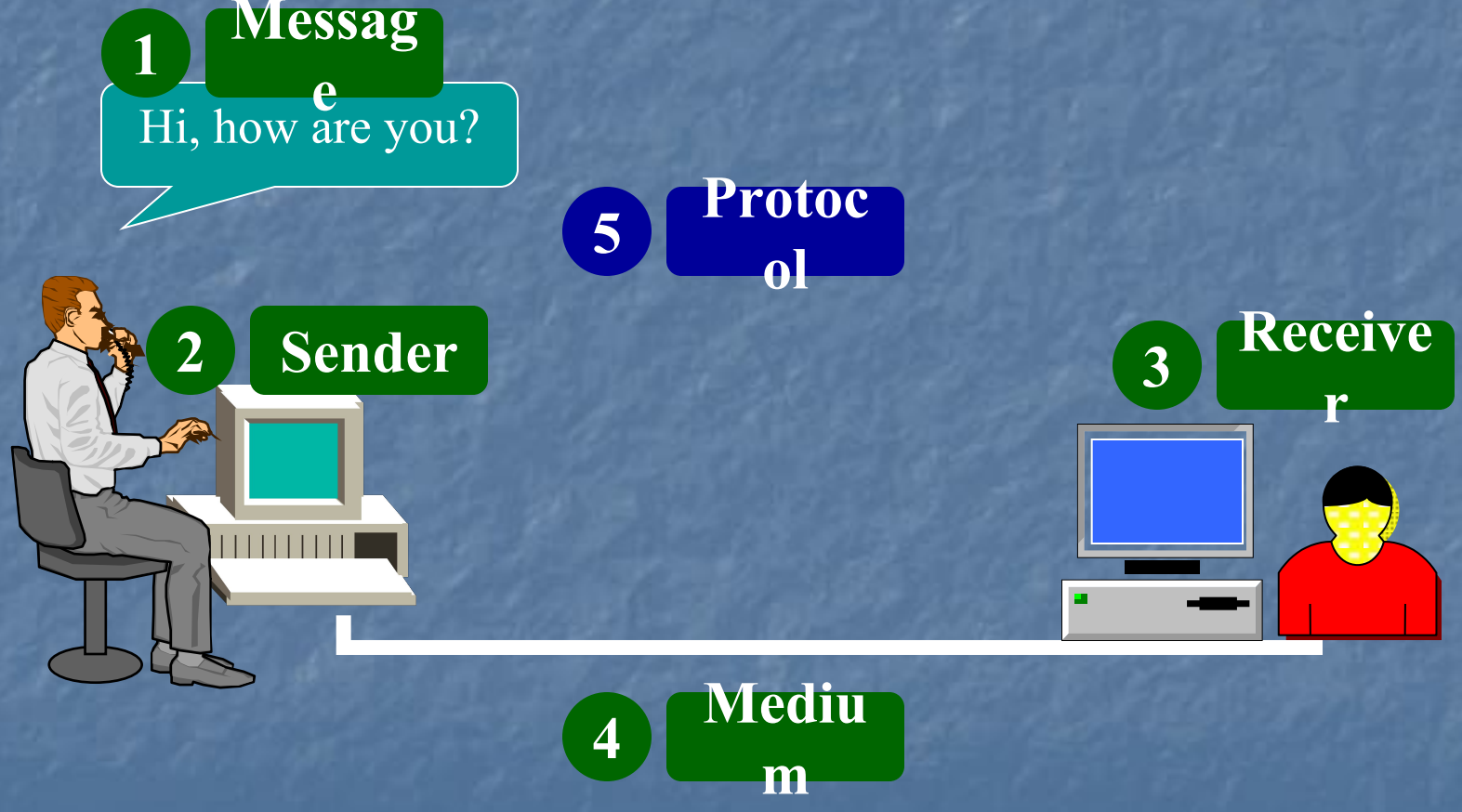
- Data Communication:

Transfer of data from one device to another via some form of transmission medium.

Data Communication



Components in Communication



Data Representation

- Numbers
  - 8/16/32 bit integers
  - floating point
- Text
  - ASCII, Unicode
- Images
  - Bit patterns, Graphics formats JPG/GIF/etc
- Audio → Samples of continuous signal
- Video → Sequence of bitmap images

Protocols and Standards

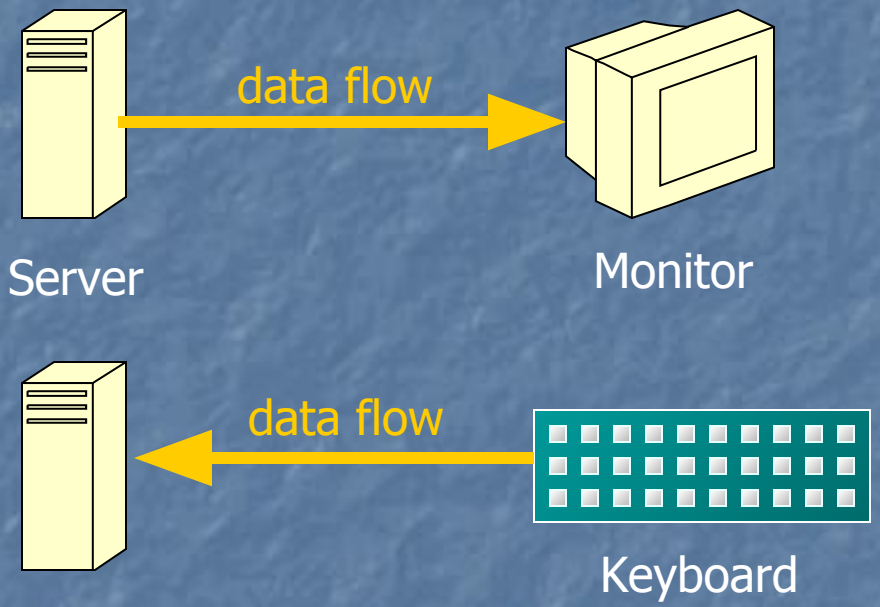
- **Protocol**
  - A set of rules governing data communications
    - **Syntax:** format of data block
    - **Semantics:** meaning of each section
    - **Timing:** speed and sequencing

Standards Organizations

- Creation Committees
  - ISO, ITU-T, ANSI, IEEE, EIA
- Forums
- Regulatory Agencies

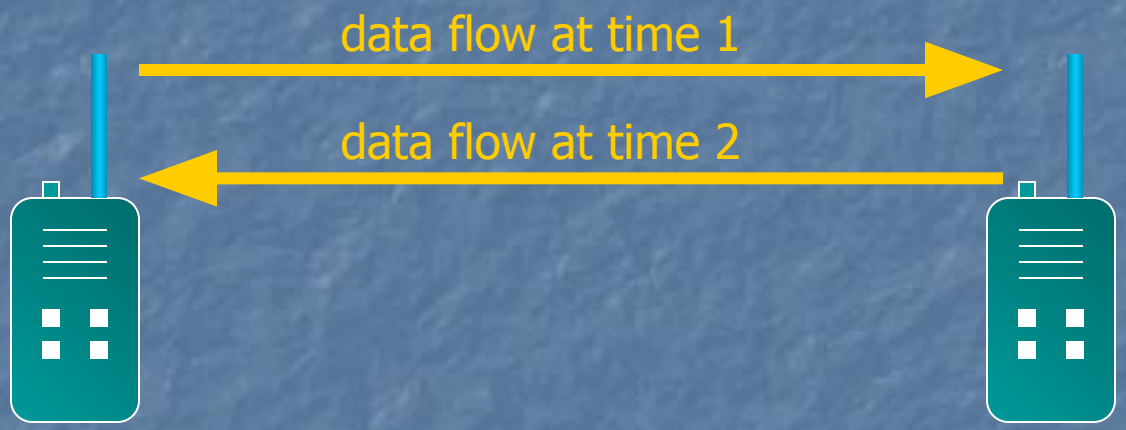
Direction of Data Flow

- **Simplex**: One direction only

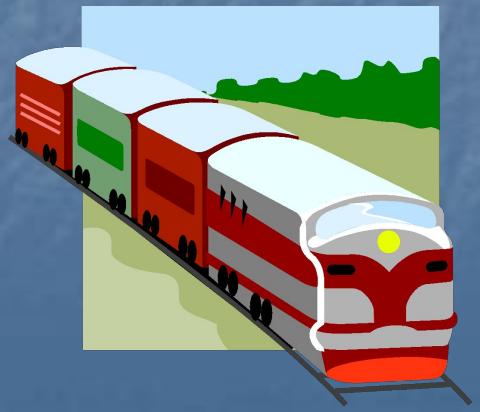


Direction of Data Flow

- **Half Duplex:** Both directions, one at a time



- E.g., walkie-talkies



Direction of Data Flow

- **Full Duplex**: Both directions simultaneously

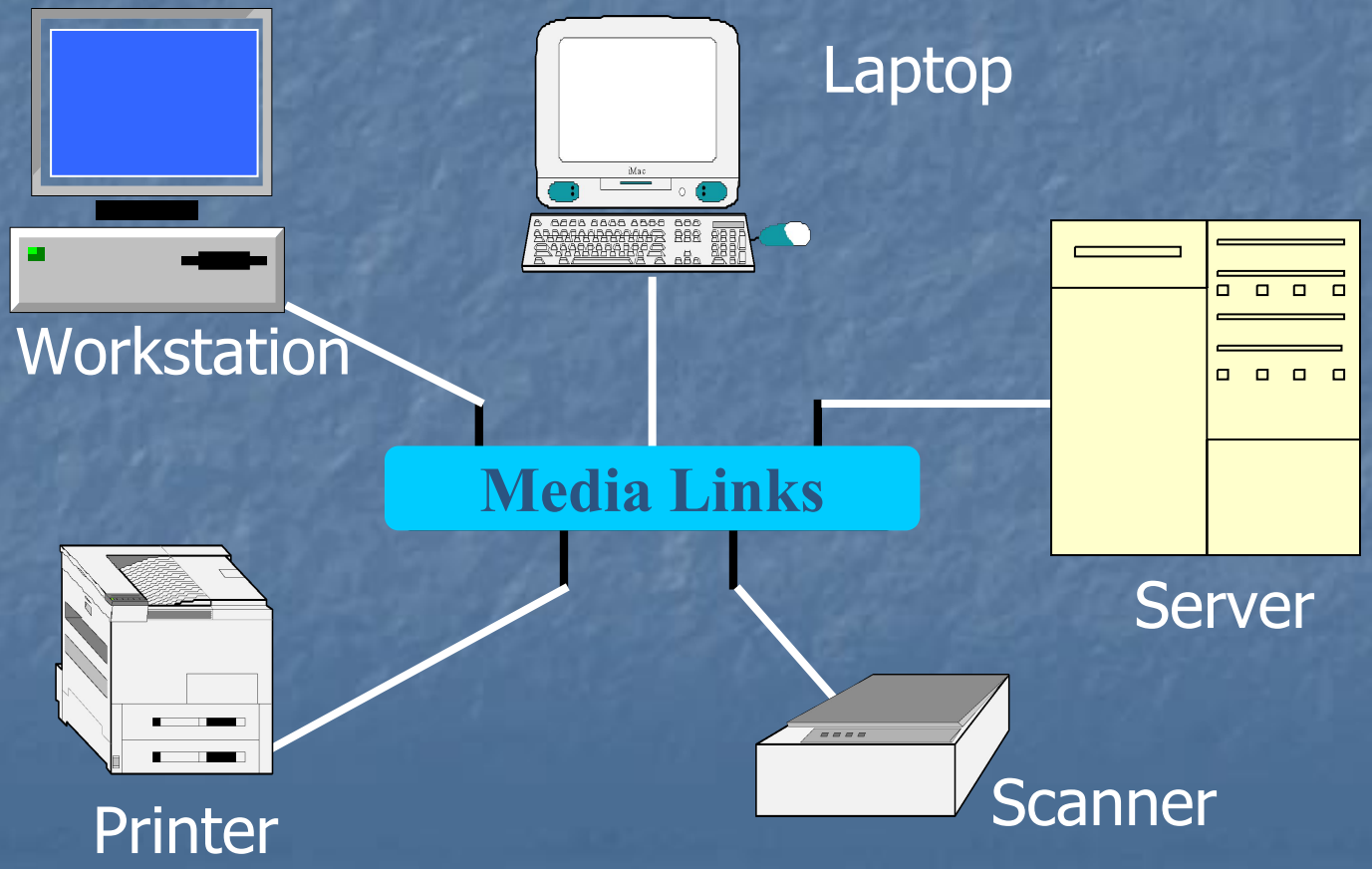


  - E.g., telephone

- Can be emulated on a single communication link using various methods

Networks

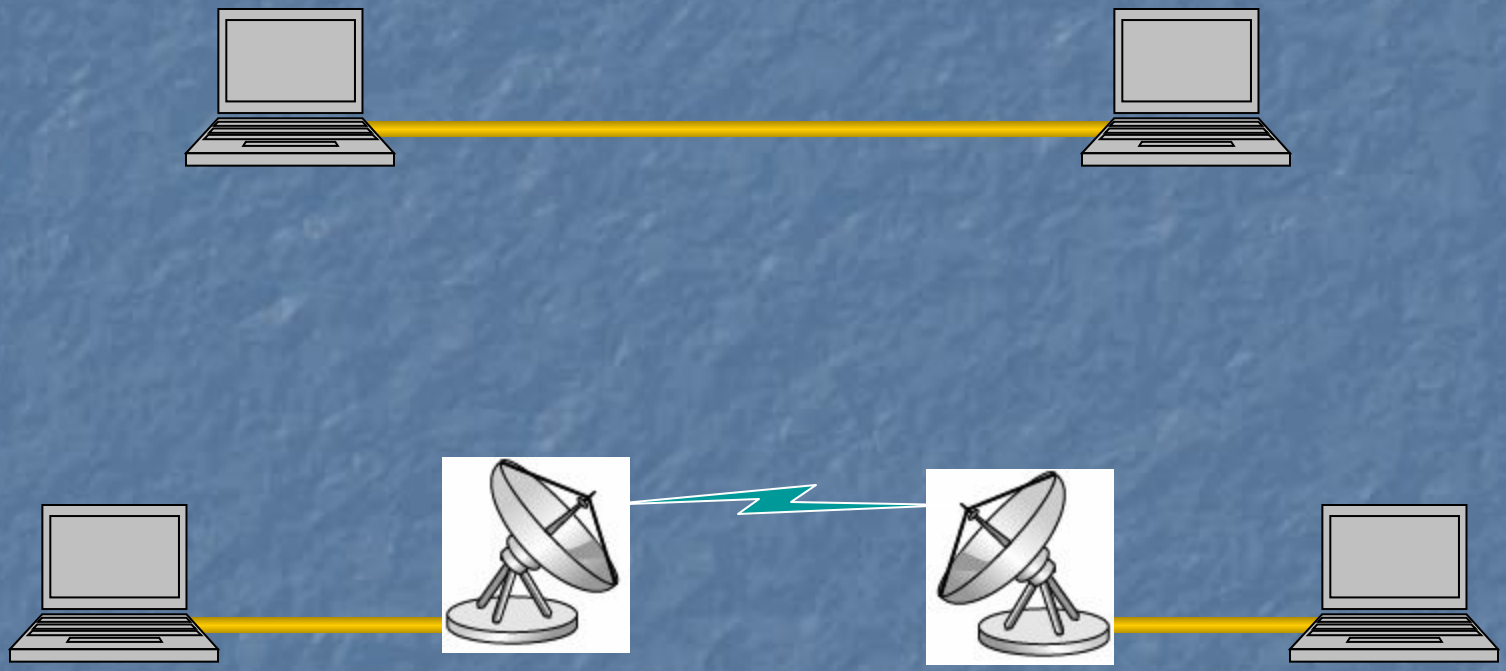
- Network: *a set of devices connected by media links*



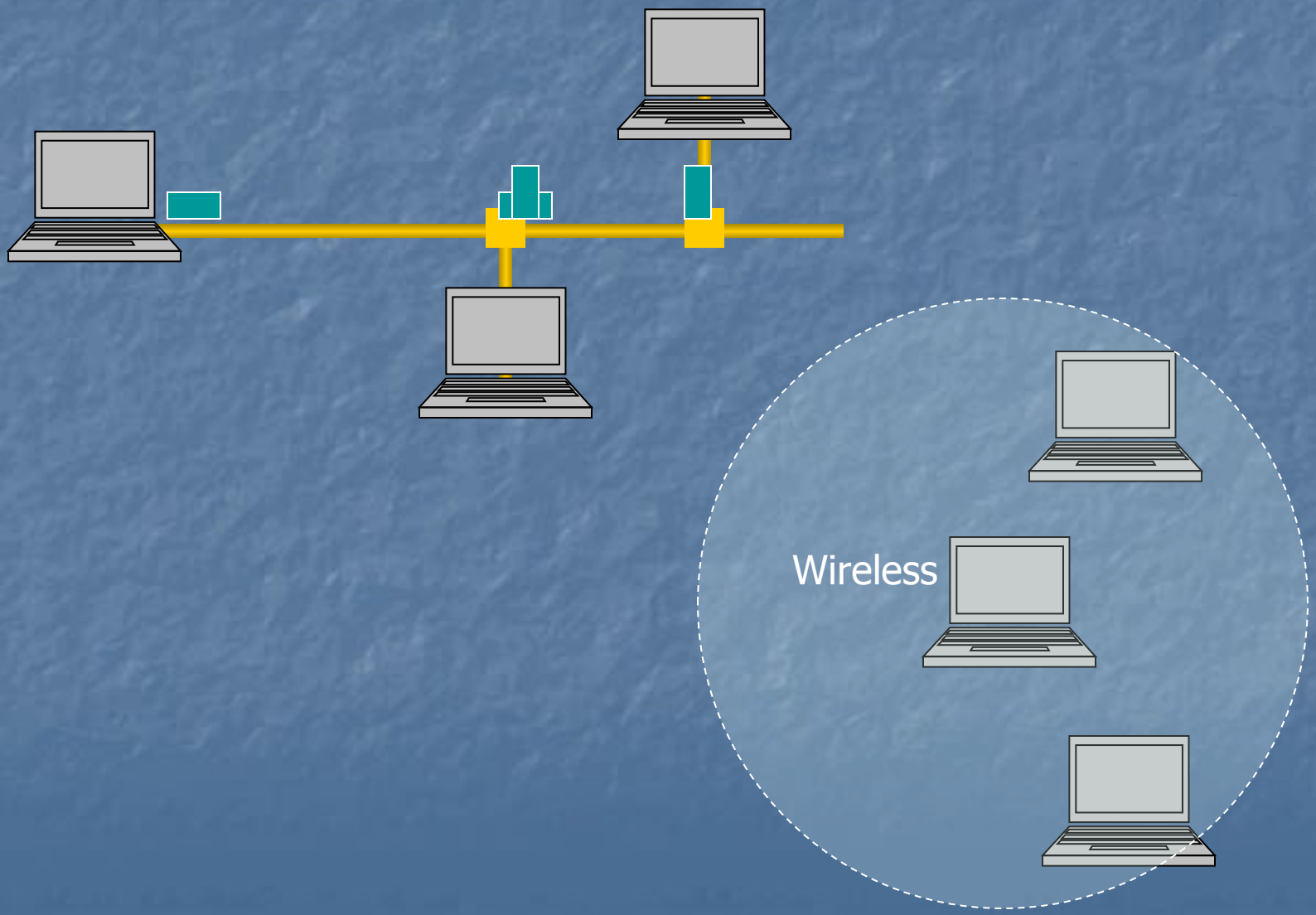
Types of Connections

- Point-to-point
- Multipoint (multidrop)

Point-To-Point Connection



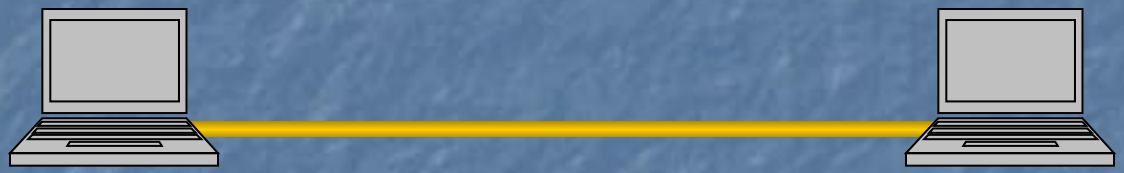
Multipoint Connection



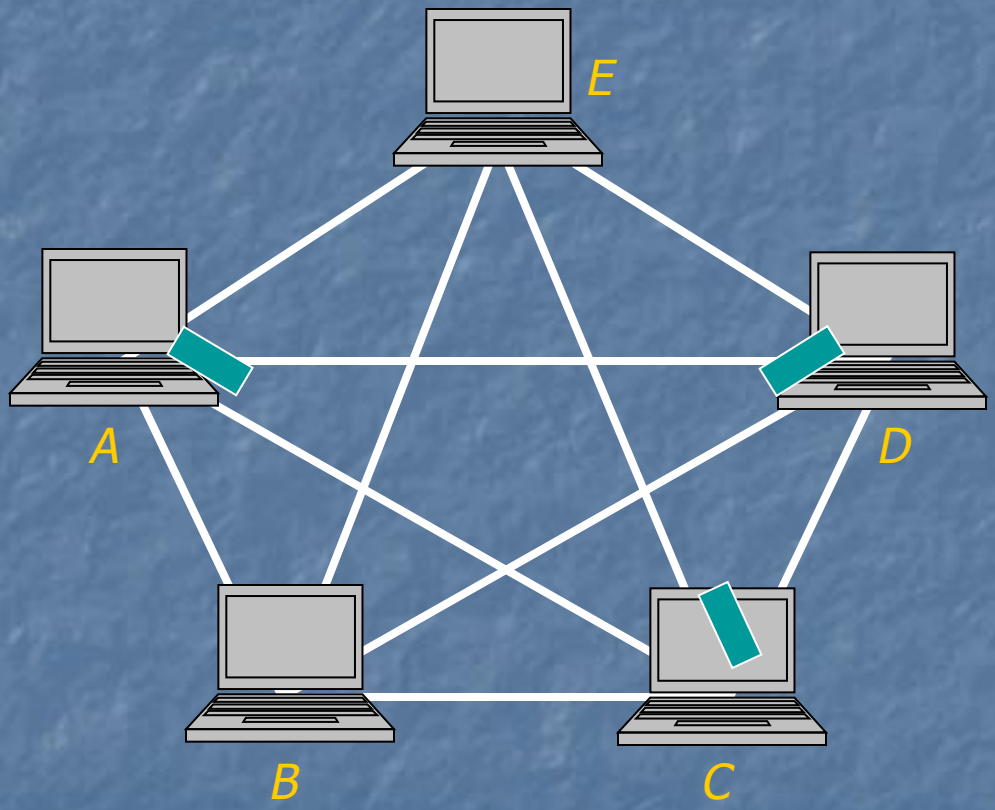
Topology

- Topology: physical or logical arrangement of devices
  - Point-to-point
  - Mesh
  - Star
  - Bus
  - Ring
  - Hybrid

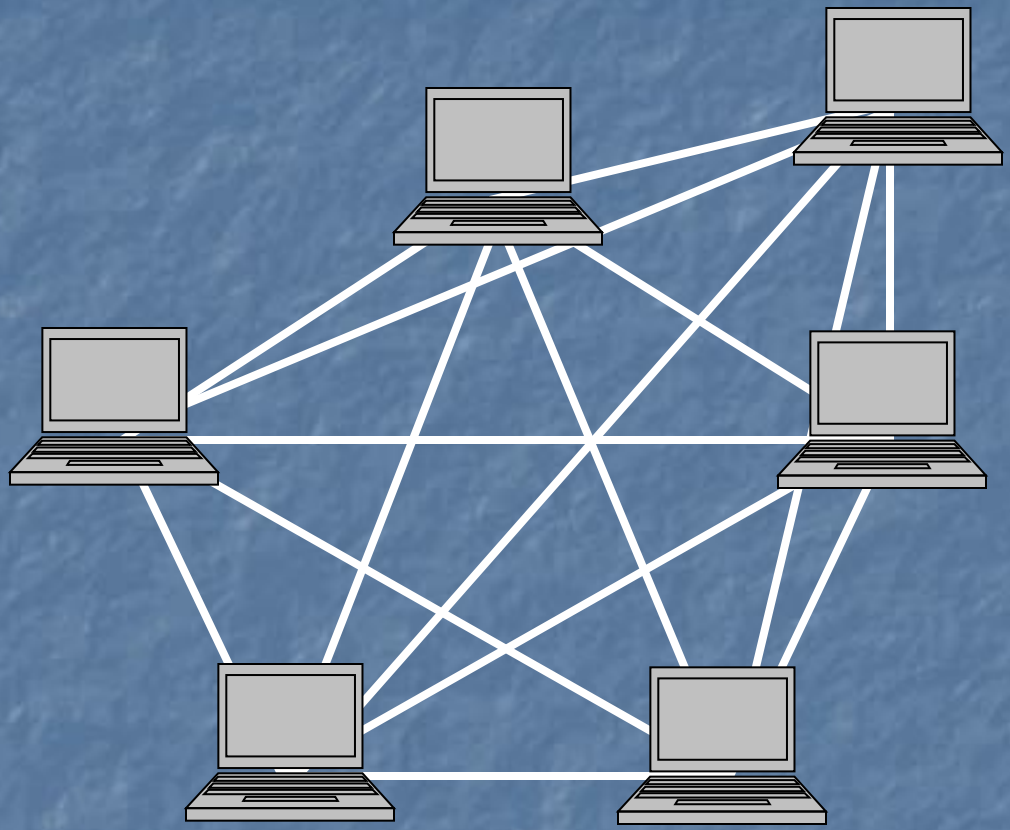
Point-to-Point Connection



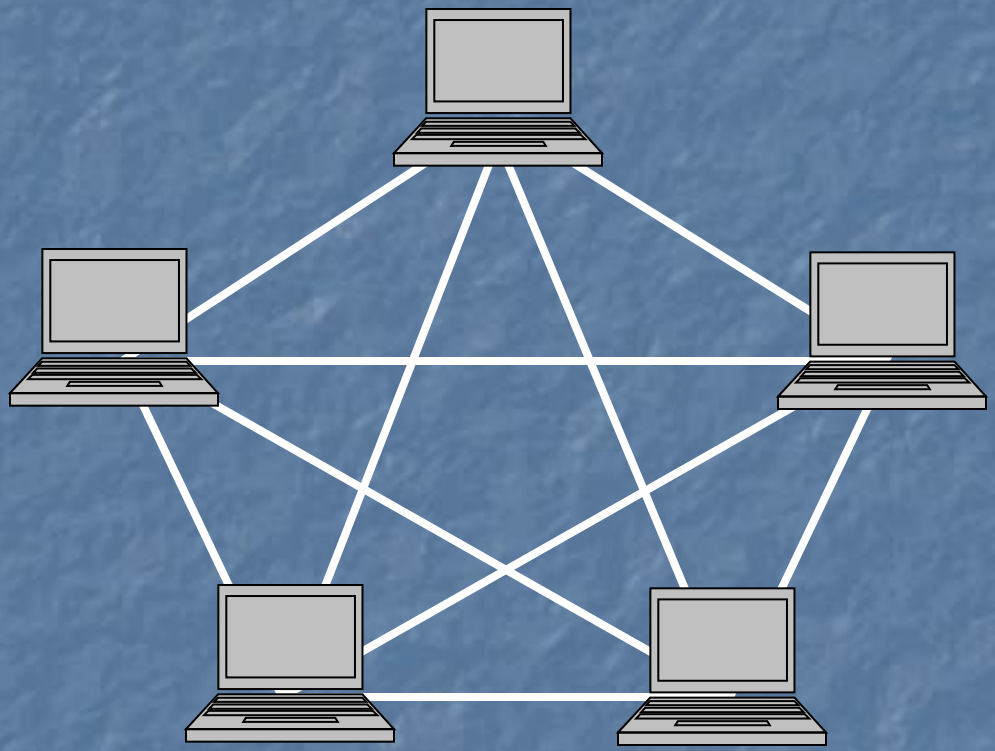
Fully Connected Mesh Topology



Fully Connected Mesh Topology

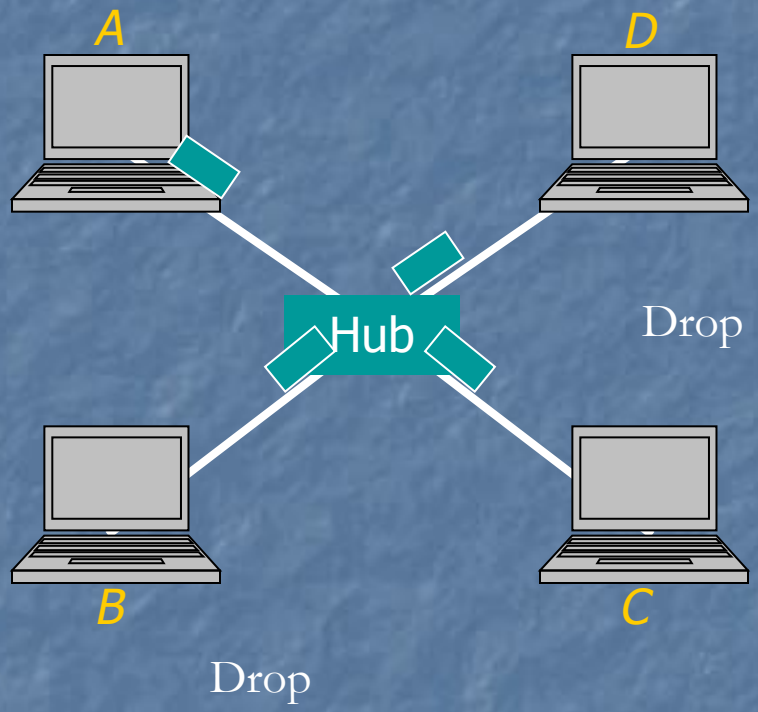


Fully Connected Mesh Topology

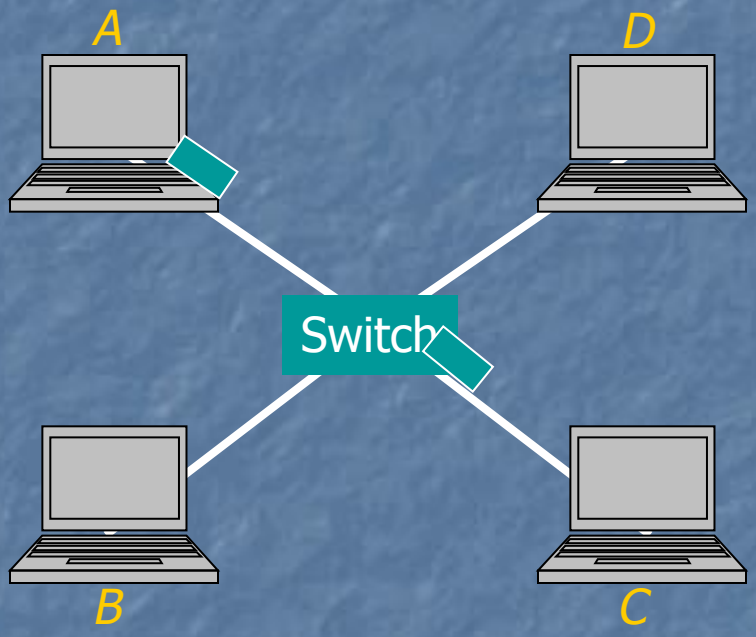


- **Pros:**
  - Dedicated links
  - Robustness
  - Privacy
  - Easy to identify fault
- **Cons:**
  - A lot of cabling
  - I/O ports
  - Difficult to move

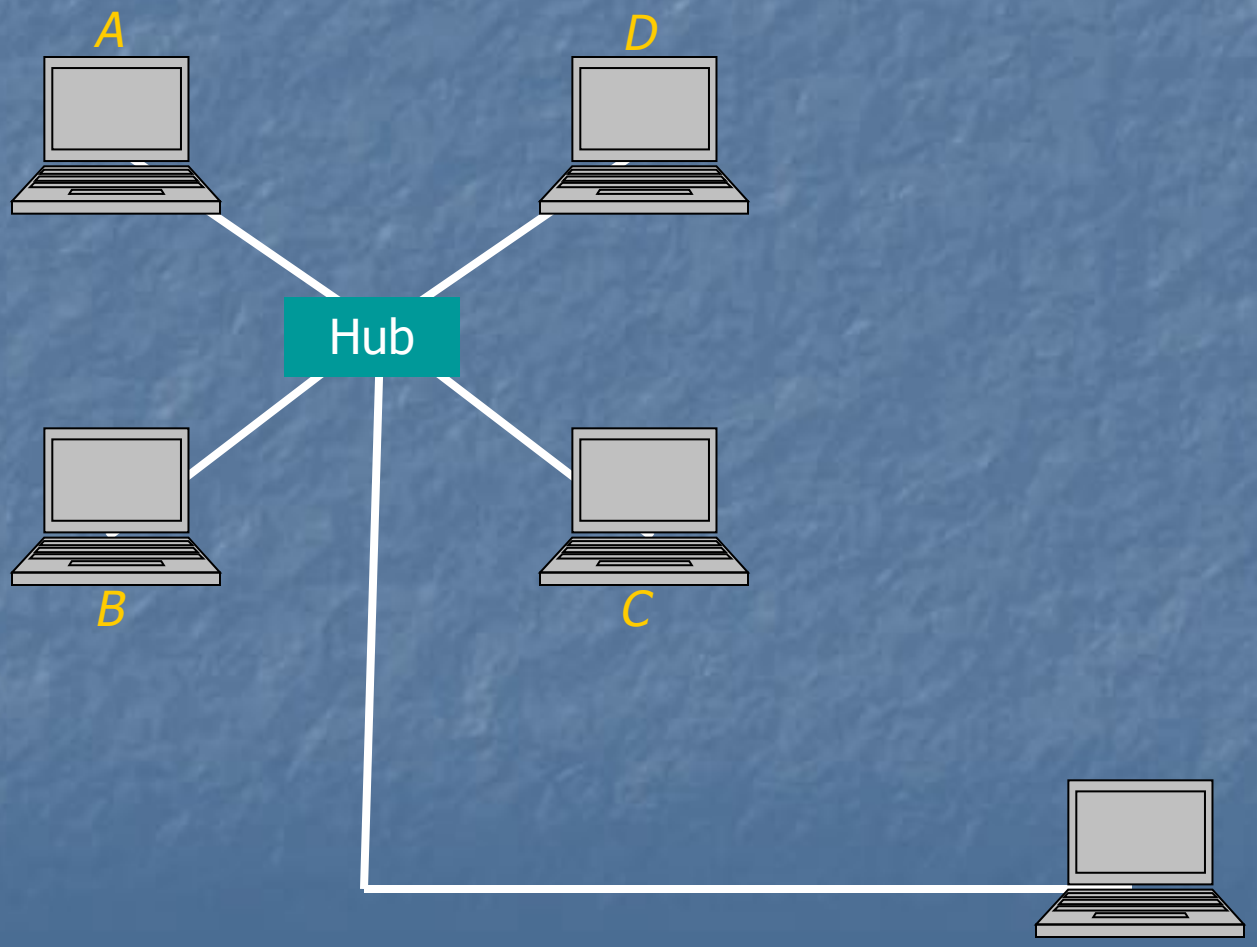
Star Topology



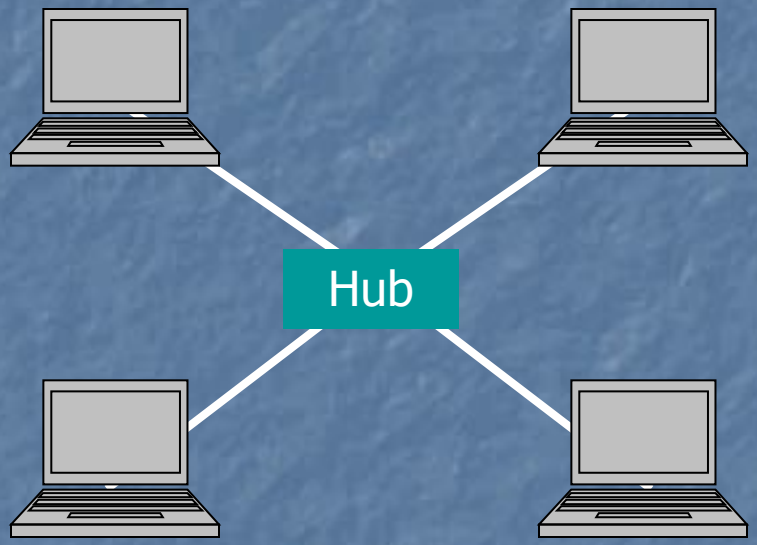
Star Topology



Star Topology

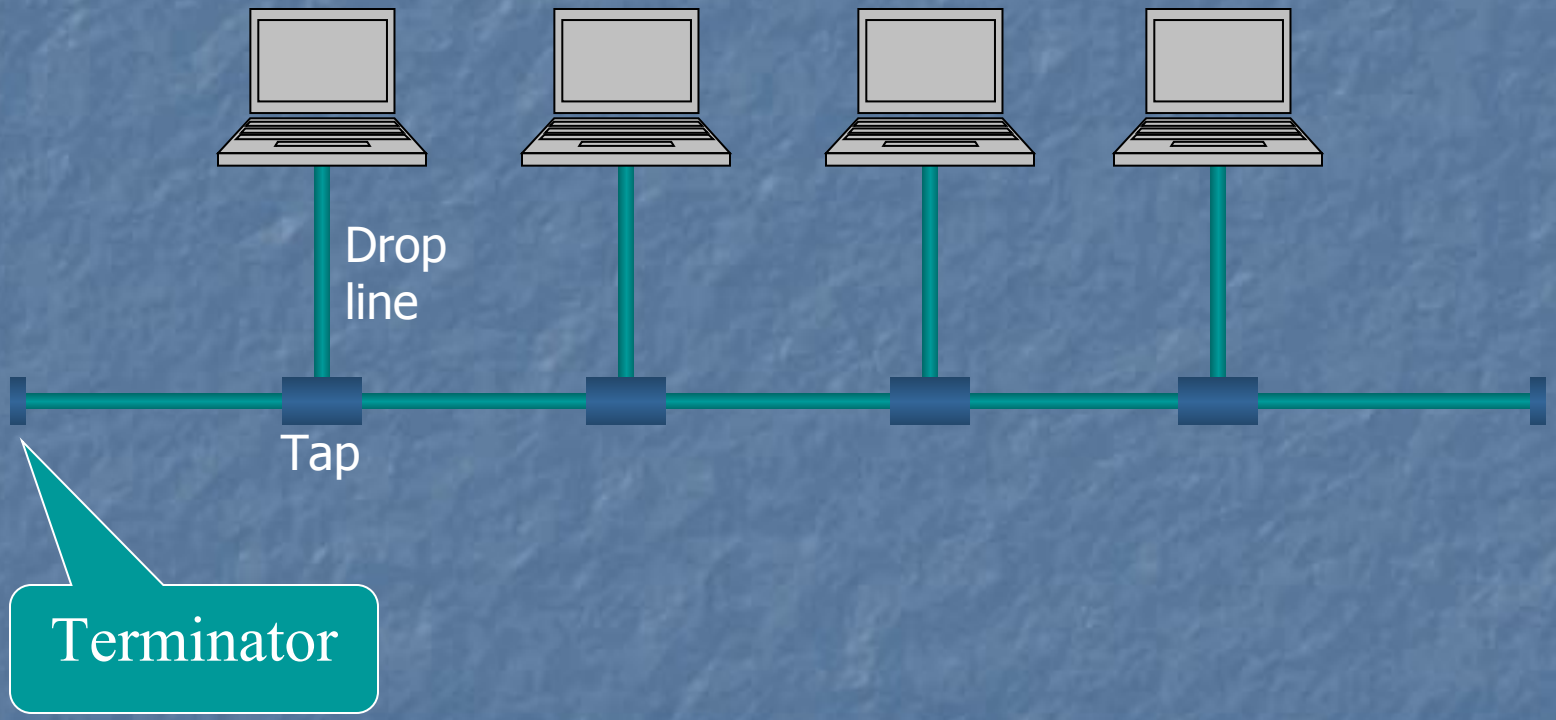


Star Topology

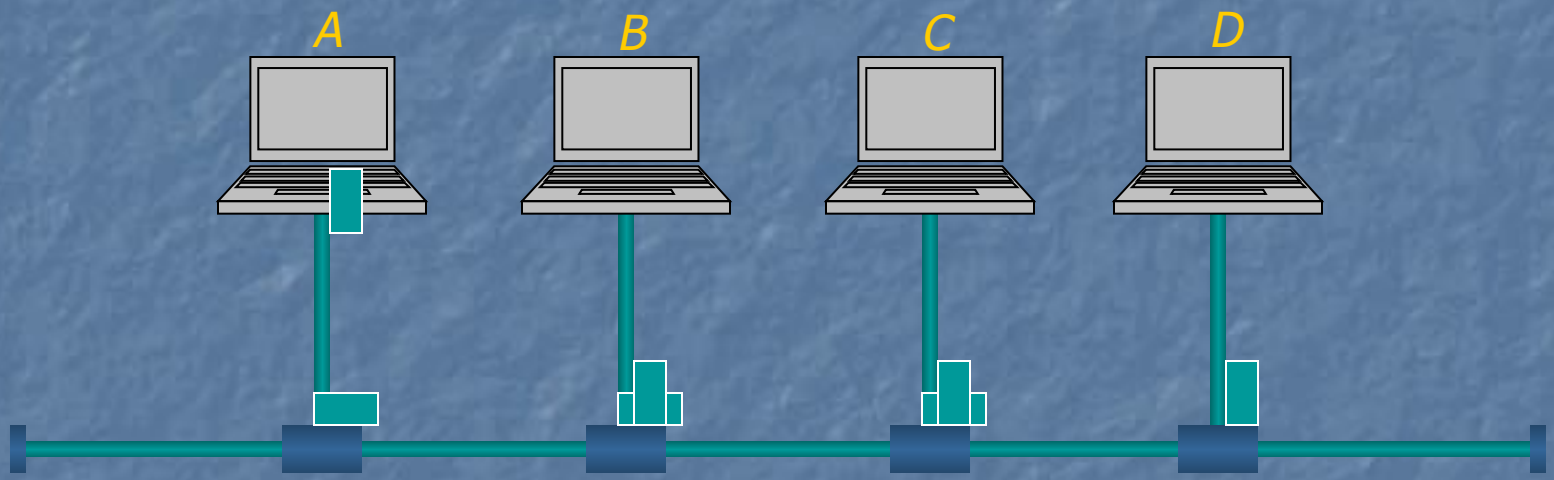


- **Pros:**
  - One I/O port per device
  - Little cabling
  - Easy to install
  - Robustness
  - Easy to identify fault
- **Cons:**
  - Single point of failure
  - More cabling still required

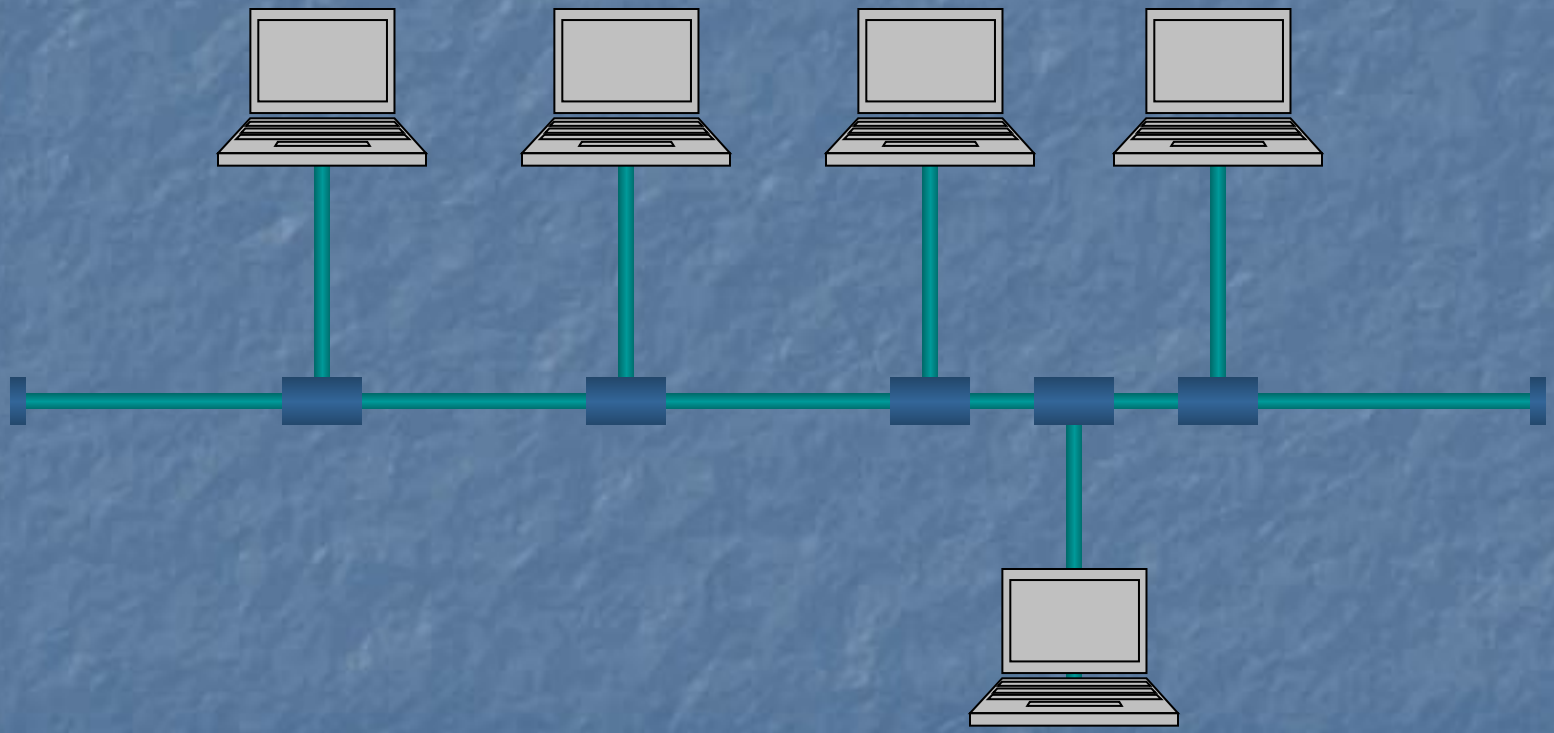
Bus Topology



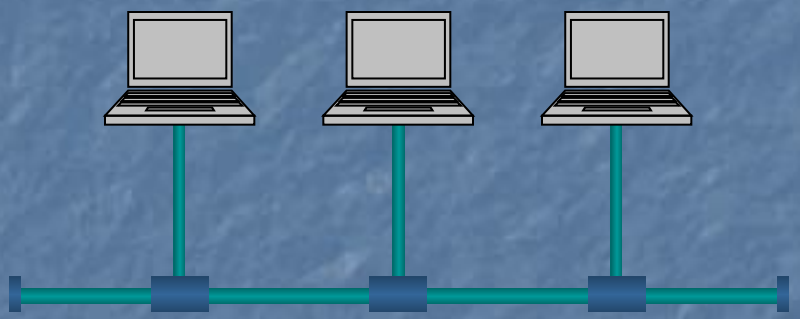
Bus Topology



Bus Topology

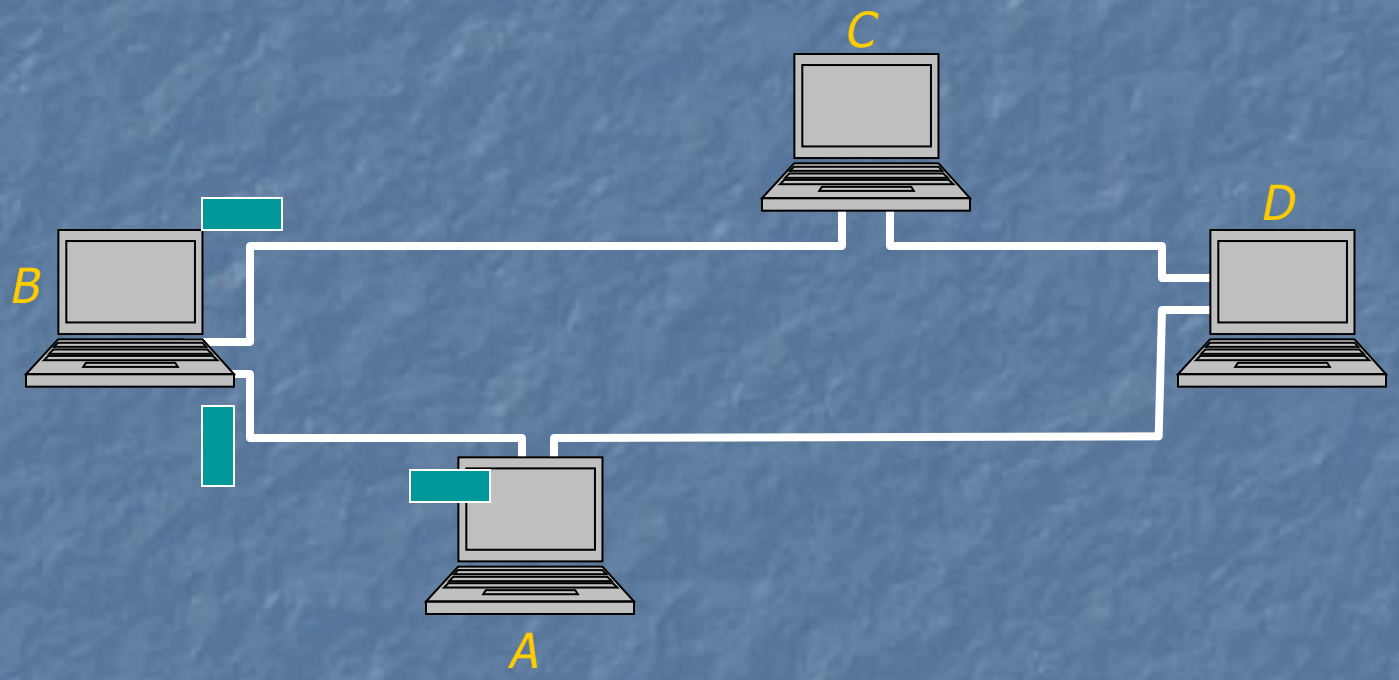


Bus Topology

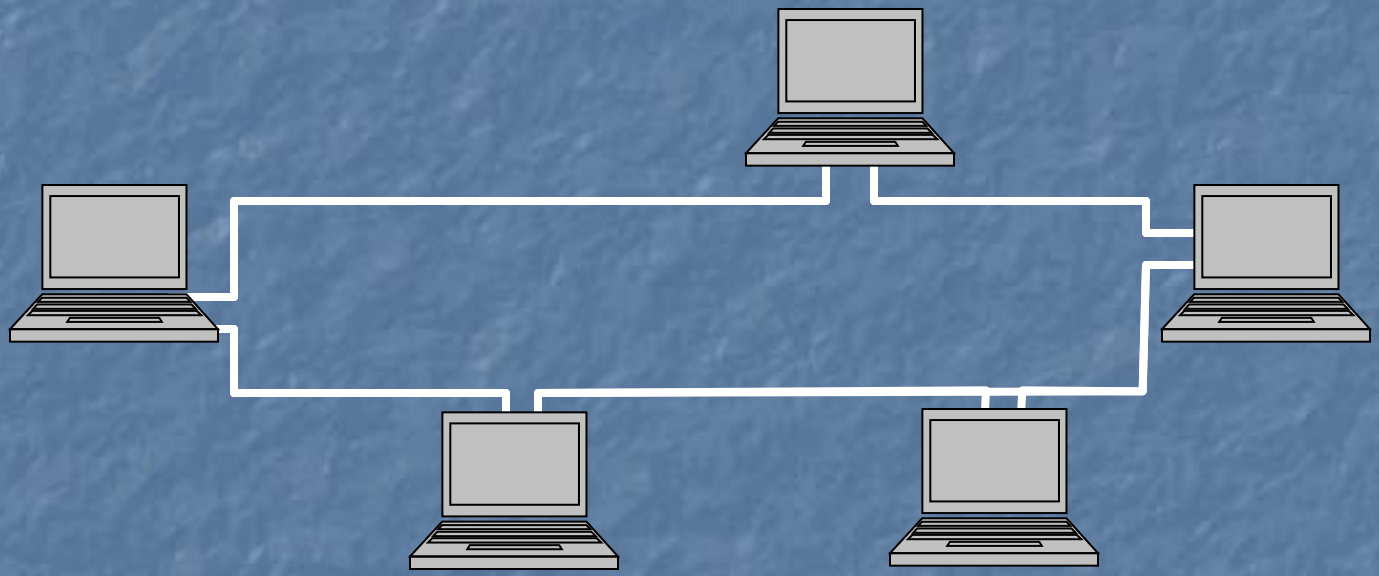


- **Pros:**
  - Little cabling
  - Easy to install
- **Cons:**
  - Difficult to modify
  - Difficult to isolate fault
  - Break in the bus cable stops all transmission

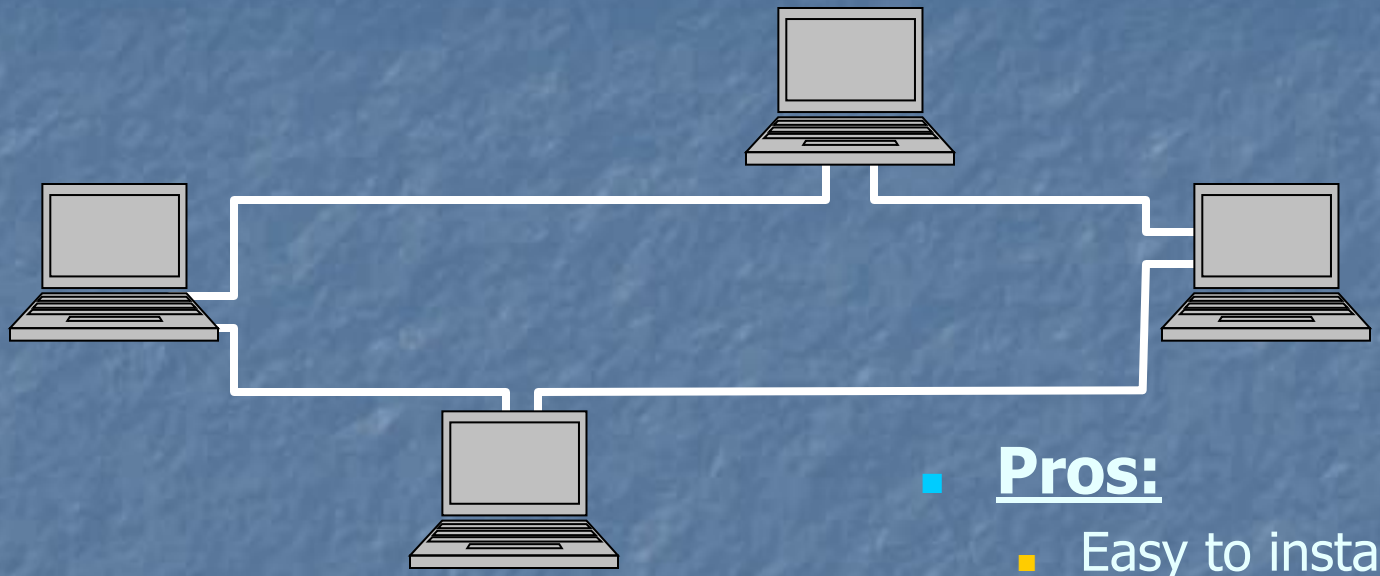
Ring Topology



Ring Topology

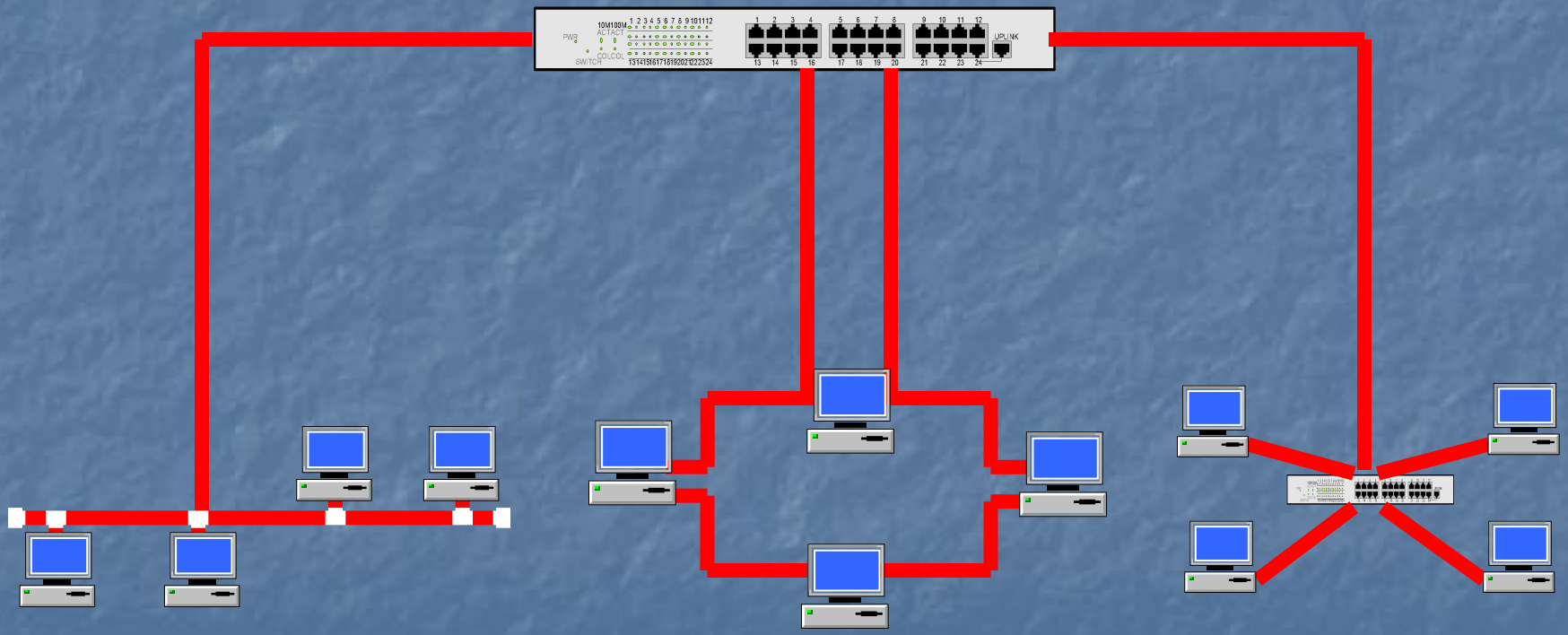


Ring Topology



- **Pros:**
  - Easy to install
  - Easy to identify fault
- **Cons:**
  - Delay in large ring
  - Break in the ring stops all transmission

Hybrid Topologies

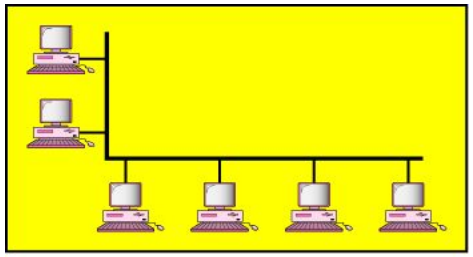


Network Categories

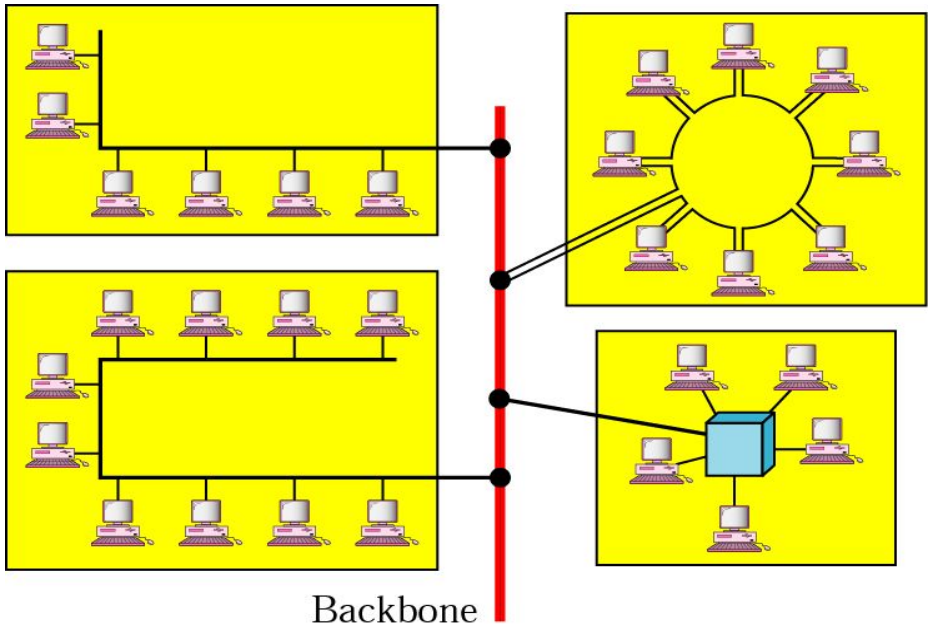
- Local Area Network (LAN)
- Wide Area Network (WAN)
- Metropolitan Area Network (MAN)

Local Area Networks

- Network in a single office, building, or campus



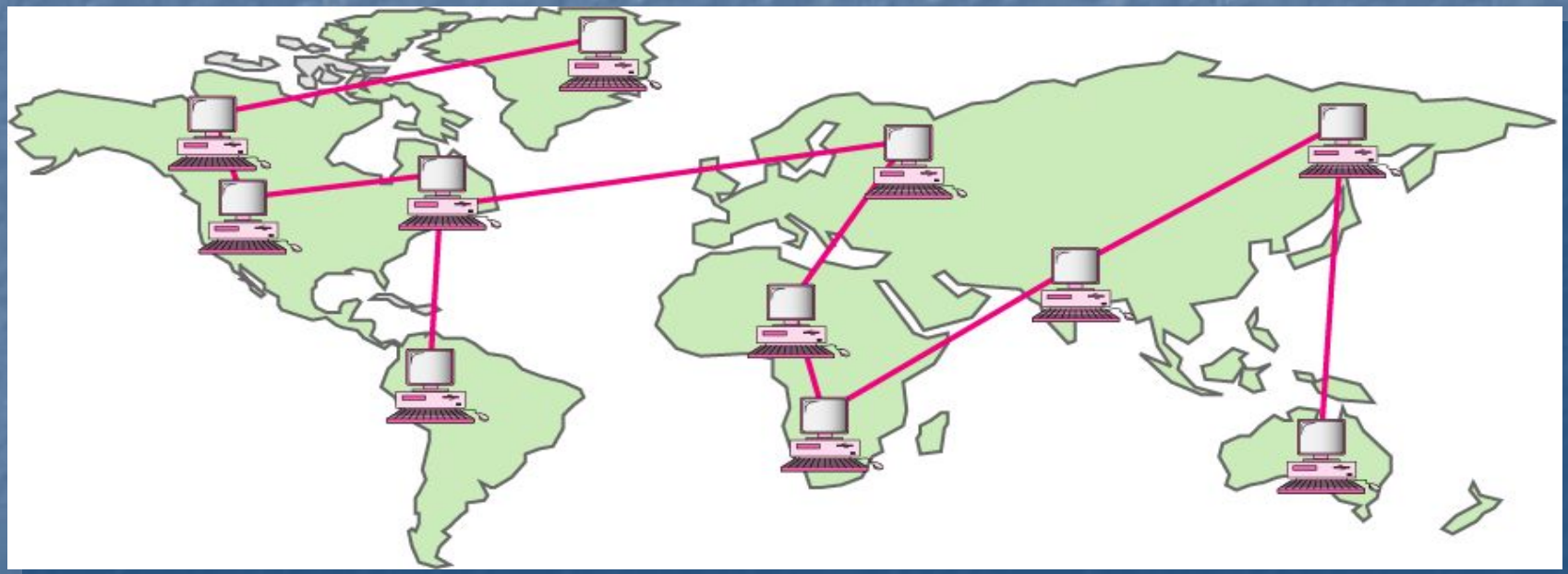
a. Single-building LAN



b. Multiple-building LAN

Wide Area Networks

- Network providing long-distance communication over a country, a continent, or the whole world



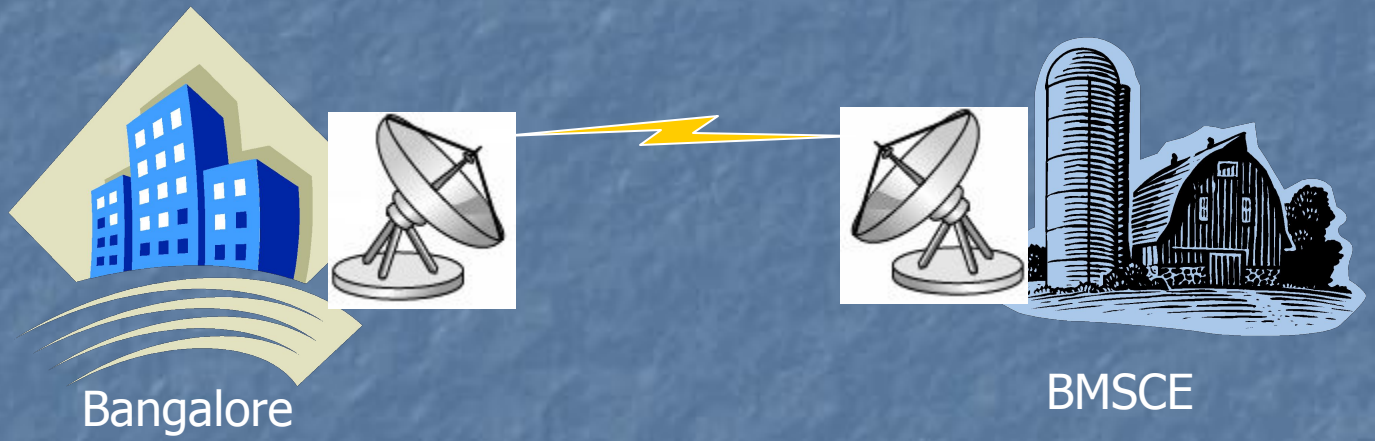
WAN Technologies

- Circuit Switching
  - Physical links are dedicated between two points
- Packet Switching
  - Data are put in packets which are stored and forwarded at each node
- Virtual Circuit Switching
  - Virtual links are established between two points

→ Frame Relay and ATM (Asynchronous Transfer Mode)

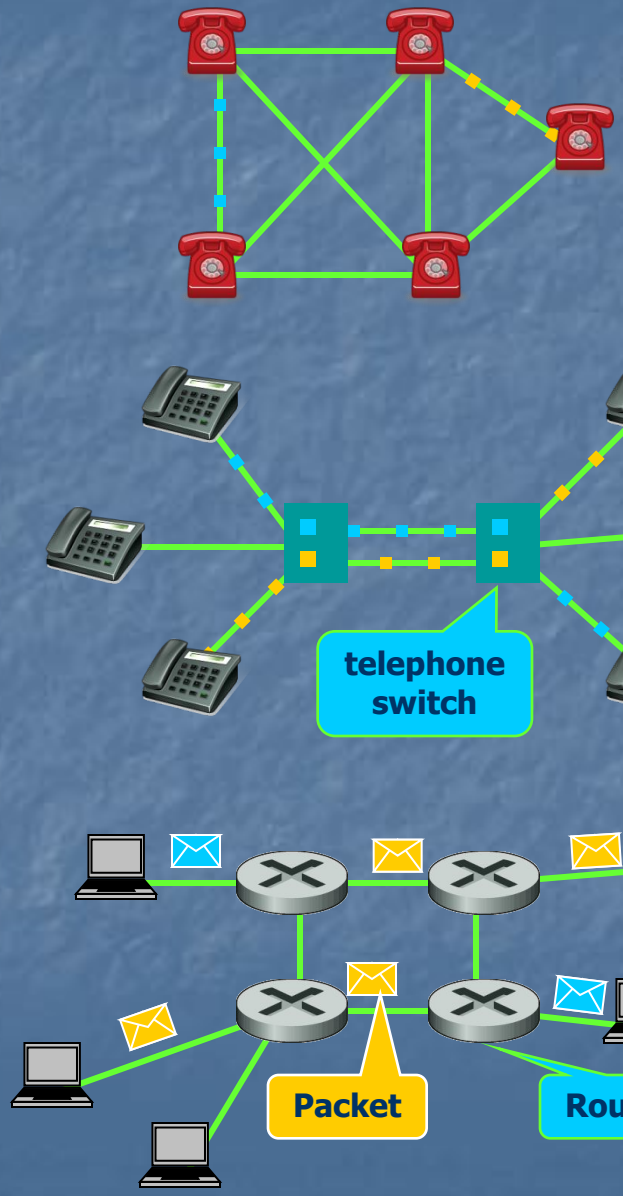
Metropolitan Area Networks

- Network extended over an entire city

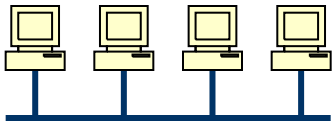


Switching

- Dedicated circuits
- Circuit switching
  - Telephone switches establish circuits for communication
- Packet switching
  - Data are put into packets
  - Each stamped with source and destination addresses
  - Routers know where to forward packets

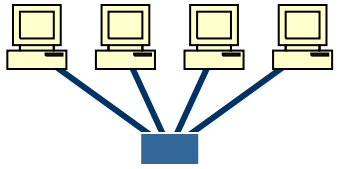


Network of Networks



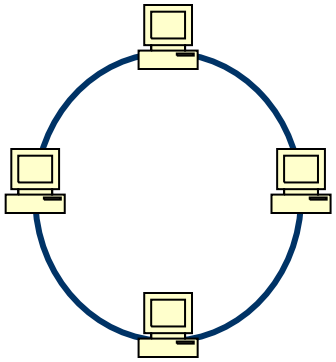
Company A





Company B

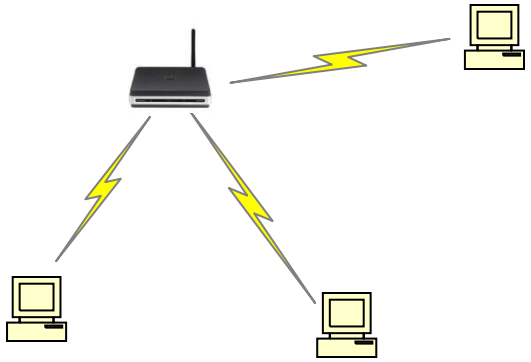




Company C



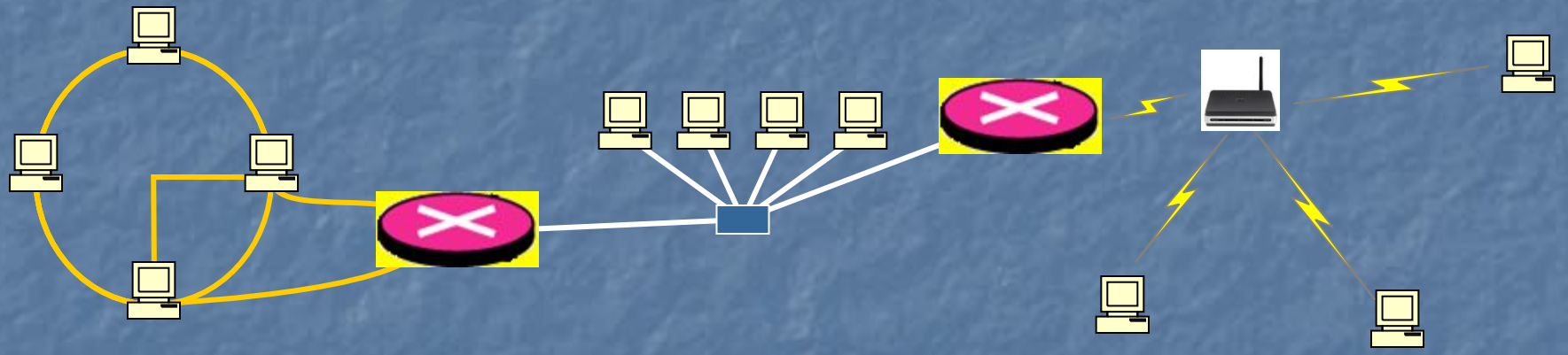




Company D

Internetworking

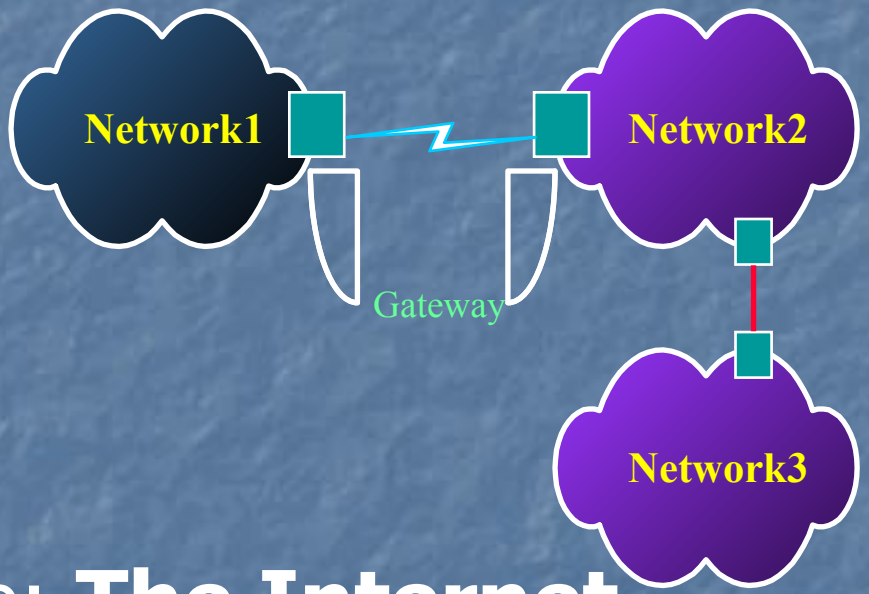
- How to allow devices from different standards to communicate
- **Gateways/routers** – devices capable of communicating in several standards



- These become "network of networks"

Internetworks

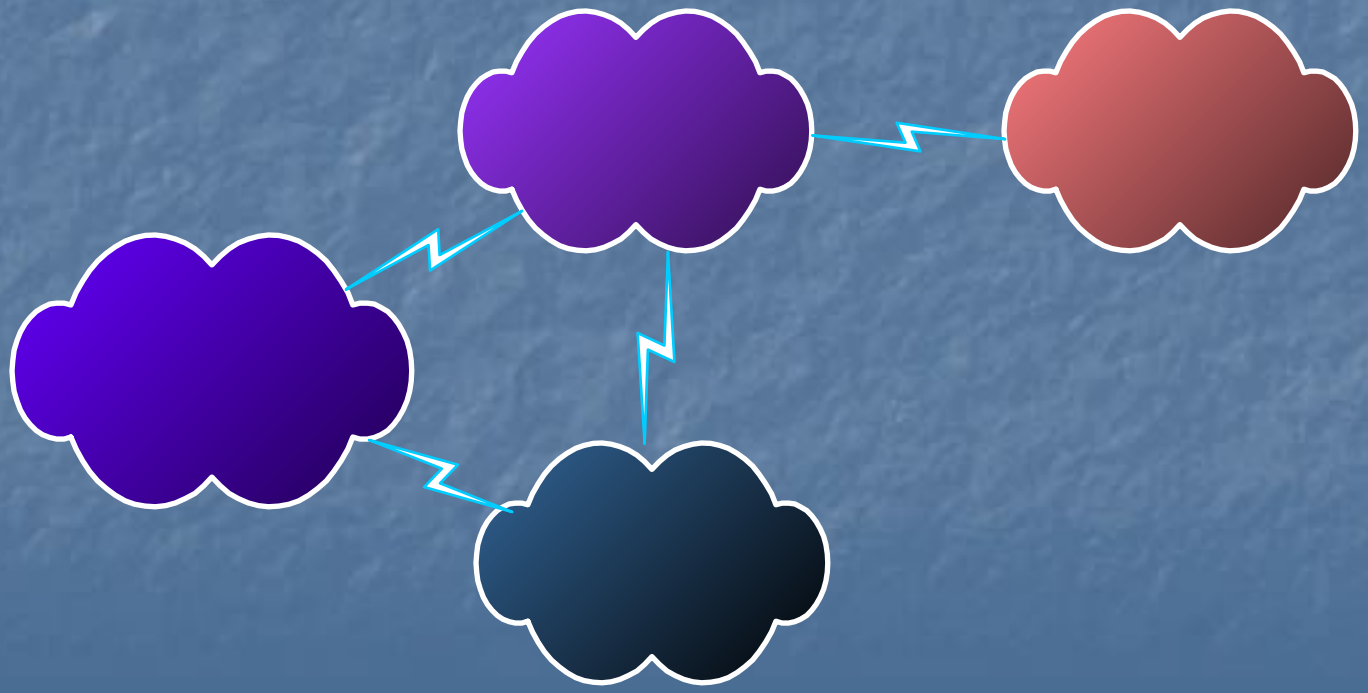
- Two or more networks connected become an *internetwork*, or *internet*



- Example: **The Internet**

The Internet

- The largest internetwork (network of networks) in the world
- Devices communicating with TCP/IP protocol suite



Summary

- Data communication
  - Protocols and standards
- Computer networks
  - Topologies
  - LAN/MAN/WAN
- Internetworks (networks of networks)